



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Outline

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Executive Summary

Objective:

- The goal is to predict the likelihood of the SpaceX Falcon 9 first stage successfully landing.
- Data is collected via SpaceX's API, web scraping, and cleaned before being used for machine learning prediction models.
- Various classification models like SVM, Decision Trees, and Logistic Regression are explored for best performance.

Introduction

- **Project Background:** SpaceX advertises Falcon 9 rocket launches with substantial cost savings compared to competitors, largely due to the ability to reuse the first stage of the rocket. Predicting if the first stage will successfully land can help competitors in making bids.
- **Problem Statement:** Can we predict whether a Falcon 9 first stage will land successfully using machine learning techniques? This prediction can offer insights into launch costs and increase competitiveness in the rocket launch market.



Section 1

Methodology

Methodology

Executive Summary

Data Collection:

- Data was gathered from the SpaceX API and through web scraping from Wikipedia to create a dataset on Falcon 9 launches. The dataset includes information such as booster version, payload mass, orbit type, launch site, and landing outcome.

Data Wrangling:

- After data collection, the data was cleaned, missing values were handled, and categorical variables (such as landing outcome) were transformed into numerical values for further analysis.

Exploratory Data Analysis (EDA):

- EDA was performed using Matplotlib and Seaborn to identify key patterns. Important variables like launch site, payload mass, and orbit type were analyzed to understand their impact on the success of Falcon 9 first stage landings.

Predictive Models:

- Several machine learning models, including Support Vector Machines (SVM), Decision Trees, and Logistic Regression, were tuned using GridSearchCV to optimize parameters and improve prediction accuracy.

Data Collection

- SpaceX API:

In this step, we utilized the SpaceX API to gather data on Falcon 9 rocket launches. The process involved making HTTP GET requests to specific API endpoints to retrieve detailed information about various aspects of the launches. The data collected includes details on the rockets, launch sites, payloads, cores, and landing outcomes.

Here's a detailed breakdown of the process:

1. Making API Requests:

Data Collection - cont.

We used the requests library in Python to make API calls to the SpaceX API endpoint <https://api.spacexdata.com/v4/launches/past>. This endpoint provides comprehensive information on all past SpaceX launches.

- The response returned by the API is in JSON format, which contains nested details about each launch.
- Extracting Key Information:
 - Rocket Information: Using the rocket ID from the JSON data, additional API requests were made to the /rockets endpoint to retrieve details about each rocket, including the booster version.
 - Launch Site Details: The launch site information (latitude, longitude, and site name) was obtained through the /launchpads endpoint.
 - Payload Information: For each launch, the payload details such as mass and orbit type were retrieved from the /payloads endpoint.
 - Core Details: The /cores endpoint provided additional data about the rocket cores, including whether the landing was successful, the landing type, and the number of previous flights for the core.

Data Collection - cont.

- Data Wrangling:
 - The raw JSON response was transformed into a structured format using Pandas, with columns such as flight number, booster version, payload mass, orbit type, and landing outcome.
 - Helper functions were created to extract and clean the data from nested fields in the JSON response. For example, we extracted the rocket's booster version, launchpad location, payload mass, and landing outcome.

Data Collection – SpaceX API

- **Key Code Snippets:**

- function like `getBoosterVersion()` was created to loop through the dataset and append the booster version to the respective entries by making API requests using the rocket ID.
- Similarly, `getLaunchSite()`, `getPayloadData()`, and `getCoreData()` functions were used to extract specific information about the launchpad, payload, and core details, respectively.
- You can find the code for the API data collection in the provided GitHub repository: <https://github.com/alkhaldieid/capston/blob/main/1-data-collection-with-api.ipynb>

[25]:

	FlightNumber	Date	BoosterVersion	PayloadMass	Orbit	LaunchSite	Outcome	Flights	GridFins	Reused	Legs	LandingPad	Block	ReusedCount	Serial
4	1	2010-06-04	Falcon 9	NaN	LEO	CCSFS SLC 40	None None	1	False	False	False	None	1.0	0	B00
5	2	2012-05-22	Falcon 9	525.0	LEO	CCSFS SLC 40	None None	1	False	False	False	None	1.0	0	B00
6	3	2013-03-01	Falcon 9	677.0	ISS	CCSFS SLC 40	None None	1	False	False	False	None	1.0	0	B00
7	4	2013-09-29	Falcon 9	500.0	PO	VAFB SLC 4E	False Ocean	1	False	False	False	None	1.0	0	B10
8	5	2013-12-03	Falcon 9	3170.0	GTO	CCSFS SLC 40	None None	1	False	False	False	None	1.0	0	B10
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
89	86	2020-09-03	Falcon 9	15600.0	VLEO	KSC LC 39A	True ASDS	2	True	True	True	5e9e3032383ecb6bb234e7ca	5.0	12	B10
90	87	2020-10-06	Falcon 9	15600.0	VLEO	KSC LC 39A	True ASDS	3	True	True	True	5e9e3032383ecb6bb234e7ca	5.0	13	B10
91	88	2020-10-18	Falcon 9	15600.0	VLEO	KSC LC 39A	True ASDS	6	True	True	True	5e9e3032383ecb6bb234e7ca	5.0	12	B10
92	89	2020-10-24	Falcon 9	15600.0	VLEO	CCSFS SLC 40	True ASDS	3	True	True	True	5e9e3032383ecb6bb234e7cc	5.0	12	B10
93	90	2020-11-05	Falcon 9	3681.0	MEO	CCSFS SLC 40	True ASDS	1	True	False	True	5e9e3032383ecb6bb234e7ca	5.0	8	B10

Data Collection - Scraping

- Web scraping was performed on Wikipedia's "List of Falcon 9 and Falcon Heavy launches" page to collect additional records of past launches. The page was accessed using the `requests` library in Python, and the HTML content was parsed using `BeautifulSoup`. We targeted the launch history table, which contained key data points such as flight number, date and time, launch site, payload, payload mass, orbit, customer, launch outcome, and booster landing outcome.

After extracting the data, we used various helper functions to clean and structure the information, handling irregularities such as missing values, inconsistent formatting, and embedded references within the table cells. The final output was a structured Pandas DataFrame containing clean and organized launch records. This dataset was combined with the data collected via the SpaceX API to ensure comprehensive coverage of all Falcon 9 launches

<https://github.com/alkhaldieid/capston/blob/main/2-data-collection-with-web-scraping.ipynb>

df											
	Flight No.	Launch site	Payload	Payload mass	Orbit	Customer	Launch outcome	Version	Booster	Booster landing	Date Time
0	1	CCAFS	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	F9 v1.07B0003.18	Failure	4 June 2010	18:45
1	2	CCAFS	Dragon	0	LEO	NASA	Success	F9 v1.07B0004.18	Failure	8 December 2010	15:43
2	3	CCAFS	Dragon	525 kg	LEO	NASA	Success	F9 v1.07B0005.18	No attempt	22 May 2012	07:44
3	4	CCAFS	SpaceX CRS-1	4,700 kg	LEO	NASA	Success	F9 v1.07B0006.18	No attempt	8 October 2012	00:35
4	5	CCAFS	SpaceX CRS-2	4,877 kg	LEO	NASA	Success	F9 v1.07B0007.18	No attempt	1 March 2013	15:10
...	...	...	...	...	...	...	...	...	...	...	...
237	117	CCSFS	Starlink	15,600 kg	LEO	SpaceX	Success	F9 B5B1051.10657	Success	9 May 2021	06:42
238	118	KSC	Starlink	~14,000 kg	LEO	SpaceX	Success	F9 B5B1058.8660	Success	15 May 2021	22:56
239	119	CCSFS	Starlink	15,600 kg	LEO	SpaceX	Success	F9 B5B1063.2665	Success	26 May 2021	18:59
240	120	KSC	SpaceX CRS-22	3,328 kg	LEO	NASA	Success	F9 B5B1067.1668	Success	3 June 2021	17:29
241	121	CCSFS	SXM-8	7,000 kg	GTO	Sirius XM	Success	F9 B5	Success	6 June 2021	04:26

Data Wrangling

1. Load and Clean Data:

- The raw SpaceX dataset was loaded from a CSV file and examined for missing values, with specific columns like **LandingPad** having around 28.88% missing values.
- Non-essential columns with high levels of missing data were excluded or handled appropriately using methods like imputation.

2. Feature Engineering:

- Key categorical variables, such as the landing outcomes (e.g., **True Ocean**, **False RTLS**), were converted into binary values for machine learning.
- A new column, **Class**, was created where **1** indicates a successful landing of the first stage, and **0** indicates failure.

3. Identify Missing Data:

- The percentage of missing values in critical columns was calculated, and appropriate strategies were applied to fill or drop these values.

4. Labeling Data:

- Launch outcomes were labeled as either successful (landing outcome = 1) or unsuccessful (landing outcome = 0). This binary classification is the target variable for the machine learning models.

5. Final Dataset:

- The final cleaned dataset includes variables such as **FlightNumber**, **LaunchSite**, **PayloadMass**, **Orbit**, and the newly engineered **Class** column.

Data Wrangling

Flowchart:

1. **Load Data** →
2. **Handle Missing Values** →
3. **Feature Engineering** (Binary Encoding, Label Creation) →
4. **Export Cleaned Data**

GitHub URL for Data Wrangling Notebook:

- [GitHub Data Wrangling Notebook](#)

EDA with Data Visualization

EDA Summary: Key Charts

- **Flight Number vs. Launch Site (Scatter Plot):**
 - **Why?:** Shows distribution of flight numbers across launch sites, highlighting frequently used sites.
- **Payload Mass vs. Launch Site (Scatter Plot):**
 - **Why?:** Analyzes how payload mass varies across launch sites, suggesting potential site-specific capabilities.
- **Launch Success by Orbit Type (Bar Chart):**
 - **Why?:** Compares success rates of launches by orbit type (LEO, GTO, etc.), offering insights into mission complexity.
- **Payload Mass vs. Orbit Type (Box Plot):**
 - **Why?:** Shows payload mass distribution for different orbits, visualizing trends and outliers.
- **Success Rate vs. Booster Version (Bar Chart):**
 - **Why?:** Analyzes landing success rates by booster version, highlighting the impact of technological improvements.

GitHub URL: [GitHub EDA with Data Visualization Notebook](#)

EDA with SQL

SQL Queries Summary:

- **Count of Total Launch Sites:**
 - Queried the unique number of launch sites used for Falcon 9 launches.
- **Total Payload Mass for NASA (CRS) Missions:**
 - Calculated the sum of the payload mass carried by NASA's CRS (Commercial Resupply Services) missions.
- **Count of Successful Landings:**
 - Retrieved the number of successful Falcon 9 first stage landings (landed on either ground pad or drone ship).
- **Mission Outcomes by Orbit:**
 - Queried the outcome of missions grouped by their target orbit (e.g., LEO, GTO, ISS).
- **Failed Landings by Launch Site:**
 - Retrieved the count of failed landings for each launch site.
- **Number of Falcon 9 Launches per Year:**
 - Calculated the total number of Falcon 9 launches grouped by year.

GitHub URL for EDA with SQL Notebook:

- [GitHub EDA with SQL Notebook](#)

Build an Interactive Map with Folium

- Folium Map Objects Summary:
 - **Markers for Launch Sites:**

Why?: Added markers at each Falcon 9 launch site to visually represent where launches took place. This helps in understanding geographical patterns related to launch success rates.

- **Circle Markers for Successful Landings:**

Why?: Circle markers were placed at locations where successful landings occurred. The radius of the circle represents the frequency of successful landings at that site, offering a quick visual insight into the success rate of each launch location.

- **Popup Labels:**

Why?: Popups were added to each marker to display additional information about the launch site, including the site name and the number of launches from that location. This provides context for users interacting with the map.

- **Polylines for Flight Paths:**

Why?: To visualize the journey of the rocket, from launch to landing, offering insights into distance and landing success by location.

https://github.com/alkhaldieid/capston/blob/main/world_map.html

Build a Dashboard with Plotly Dash

Success Rate by Launch Site (Bar Chart):

- **Why?:** This bar chart displays the success rate of Falcon 9 landings for each launch site. It provides a clear comparison of how each site has performed over time, allowing users to quickly identify the most successful launch sites.

Payload vs. Success Scatter Plot:

- **Why?:** This scatter plot shows the relationship between payload mass and landing success. Users can filter by launch site and orbit type to explore how different payload weights influence the likelihood of a successful landing.

Payload Mass Range Slider:

- **Why?:** A range slider was added to allow users to dynamically filter launches based on the payload mass. This interactivity helps users explore how different payload sizes impact landing success without being overwhelmed by too much data.

Success-Pie Chart by Orbit Type:

- **Why?:** The pie chart breaks down the landing success rates based on the type of orbit (LEO, GTO, etc.). This visual makes it easy to see which orbits have the highest and lowest success rates, offering insights into how mission parameters affect landing outcomes.

Dropdown Filters (Launch Site and Orbit Type):

- **Why?:** Dropdown filters were added to allow users to select specific launch sites or orbit types, dynamically updating the charts to reflect the filtered data. This interaction enhances the exploration of different mission outcomes based on launch site and orbit.

GitHub URL for Completed Plotly Dash Lab:

- [GitHub Plotly Dash Lab](#)

Predictive Analysis (Classification)

- **Models Chosen:**
 - Logistic Regression
 - Support Vector Machine (SVM)
 - Decision Trees
 - K-Nearest Neighbors (KNN)
- **Why?:** These models are widely used for classification tasks and offer a variety of approaches, from linear (Logistic Regression) to non-linear (SVM, Decision Trees, KNN).

Predictive Analysis (Classification)

Data Cleaning:

- Encoding categorical variables (e.g., landing outcome)
- Handling missing values

Feature Selection:

- Focused on key variables: Payload Mass, Orbit Type, Booster Version

Data Scaling:

- Applied scaling techniques to ensure models perform well, particularly SVM and KNN.

Predictive Analysis (Classification)

Tuning Process:

- Used **GridSearchCV** to optimize parameters.

Parameters Tuned:

- Logistic Regression: **C** (Regularization)
- SVM: **C**, **Kernel**, **Gamma**
- Decision Trees: **Max Depth**, **Min Samples Split**
- KNN: **Number of Neighbors**

Cross-validation:

- Used 10-fold cross-validation to validate models.

Predictive Analysis (Classification)

Model Evaluation

- **Evaluation Metrics:**
 - Accuracy, Precision, Recall
 - Confusion Matrices
- **Best Results:**
 - **Decision Tree:** Validation accuracy of 90.36%
 - **SVM:** Validation accuracy of 87.5%

Best Performing Model

- **Best Model: Decision Tree**
 - Final Test Accuracy: 83.33%
- **Why?:**
 - Strikes a good balance between interpretability and performance.

Results

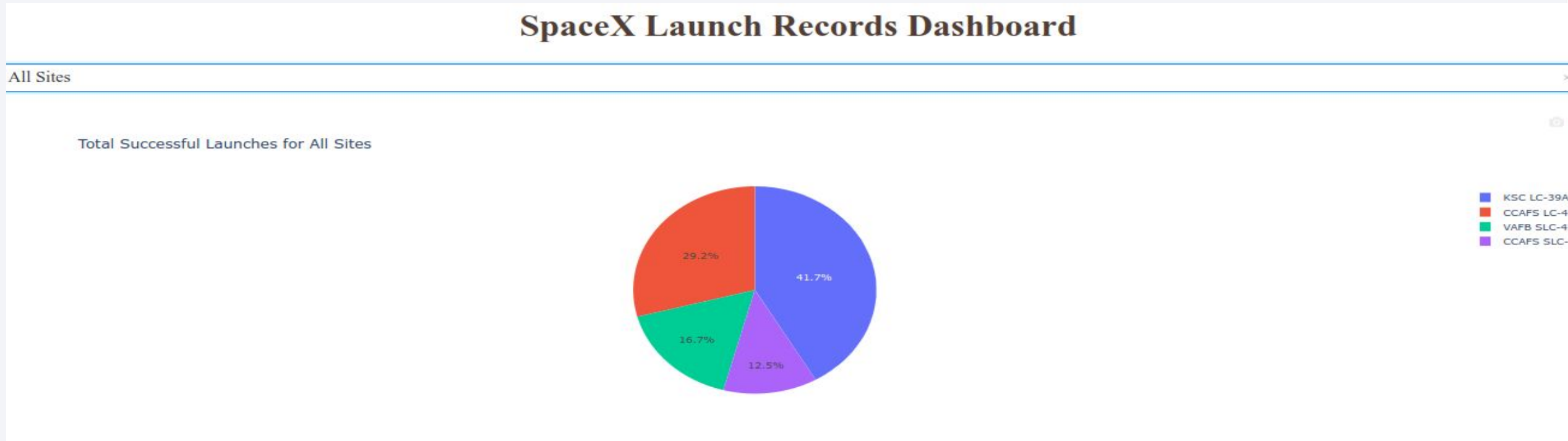
Exploratory Data Analysis (EDA) Results

- **Key Findings:**
 - **Flight Number vs. Launch Site:**
 - Most launches occurred at CCAFS SLC 40, followed by KSC LC 39A.
 - **Payload Mass vs. Launch Site:**
 - Heavier payloads were primarily launched from CCAFS SLC 40, suggesting site specialization.
 - **Launch Success by Orbit Type:**
 - LEO (Low Earth Orbit) and GTO (Geostationary Transfer Orbit) had higher success rates compared to other orbits.
 - **Payload Mass vs. Orbit Type:**
 - GTO missions generally carry heavier payloads, while LEO missions handle smaller payloads.
 - **Success Rate vs. Booster Version:**
 - Newer booster versions showed higher success rates, indicating improvement in SpaceX's technology over time.

Results

Interactive Analytics Demo (Screenshots)

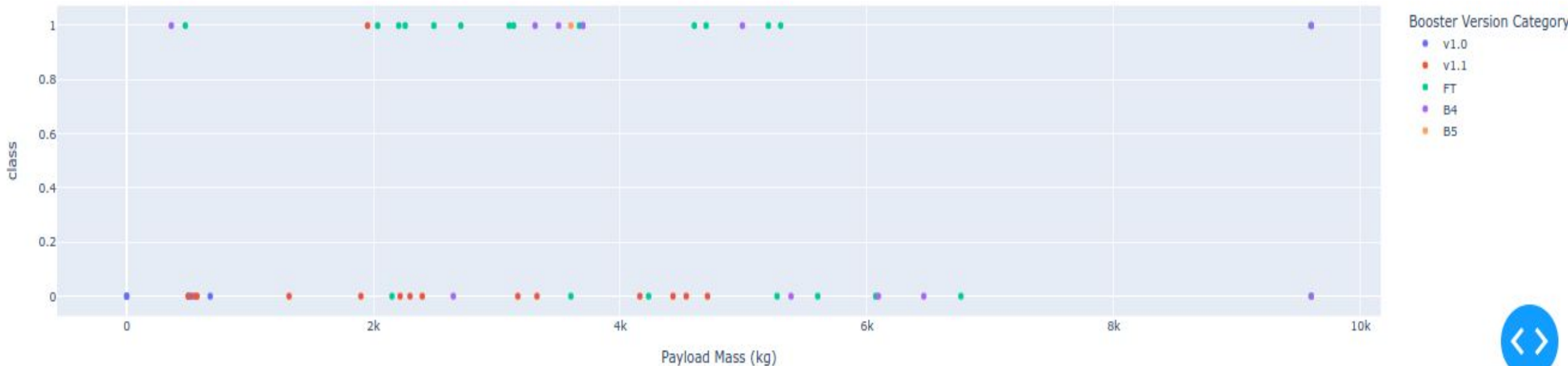
- **Screenshot 1:** Success Rate by Launch Site (Bar Chart)
 - **Explanation:** Visualizes the success rate for each SpaceX launch site.



Results

- **Screenshot 2: Payload vs. Success Scatter Plot**
 - i. **Explanation:** Allows exploration of how payload mass affects landing success.

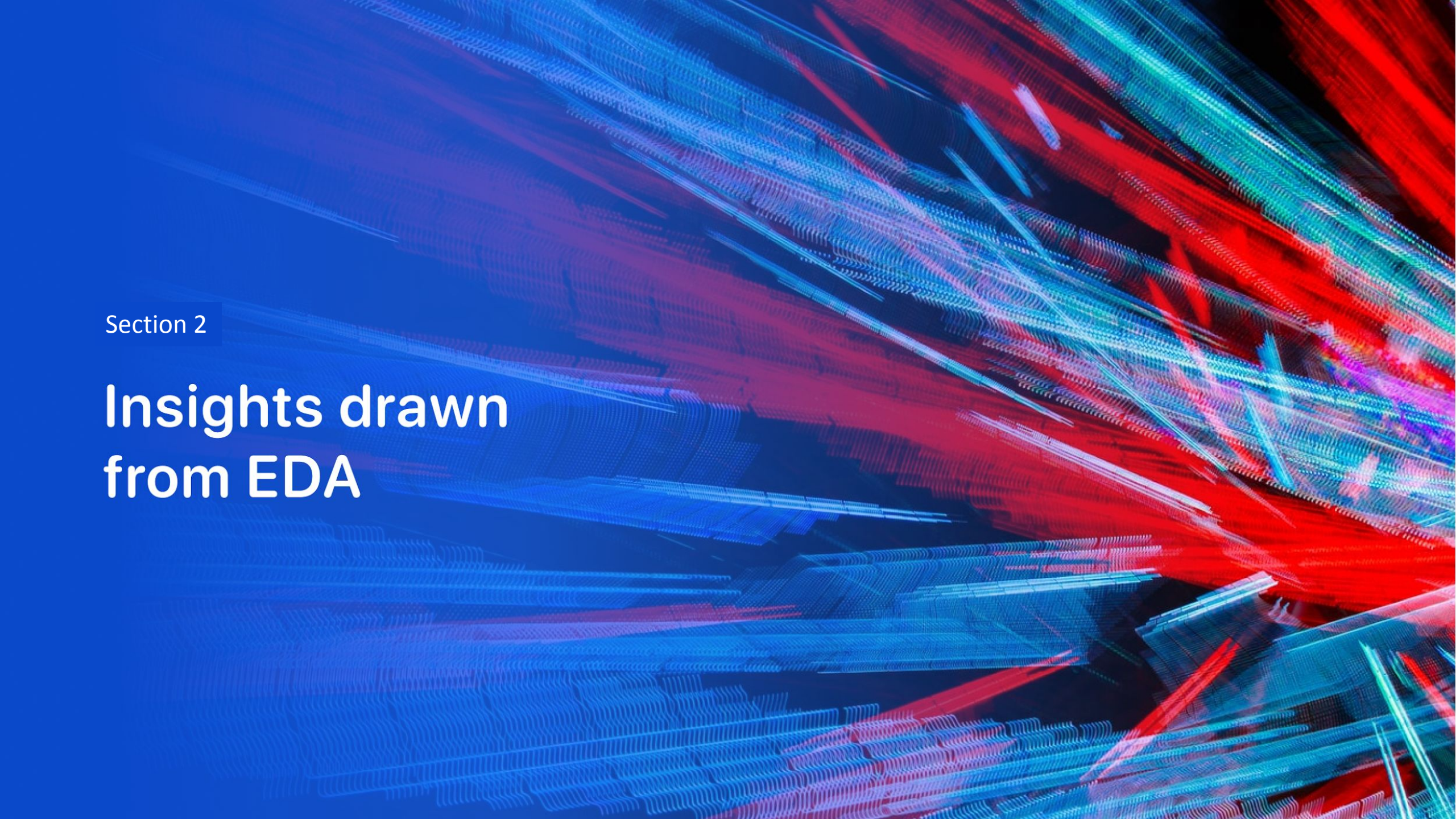
Payload vs Success for All Sites



Results

- **Screenshot 3: Payload Mass Range Slider**
 - i. **Explanation:** Interactive slider for filtering data based on payload mass to explore trends in success rates.



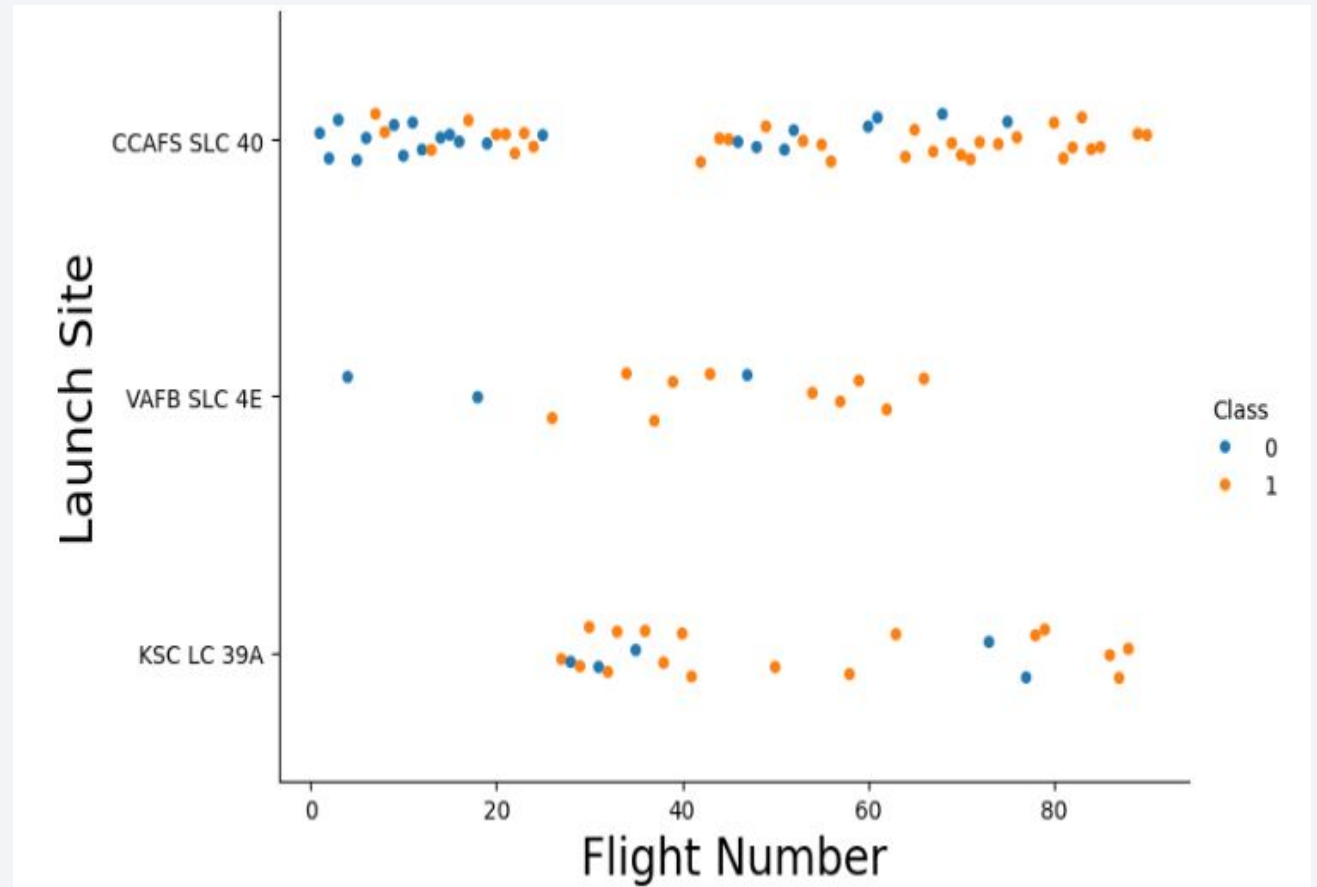
The background of the slide is an abstract composition. It features a solid blue area on the left side, which transitions into a complex pattern of diagonal streaks and a grid-like texture on the right. The streaks are primarily in shades of blue and red, with some green and purple accents. The overall effect is dynamic and modern, suggesting a digital or data-driven theme.

Section 2

Insights drawn from EDA

Flight Number vs. Launch Site

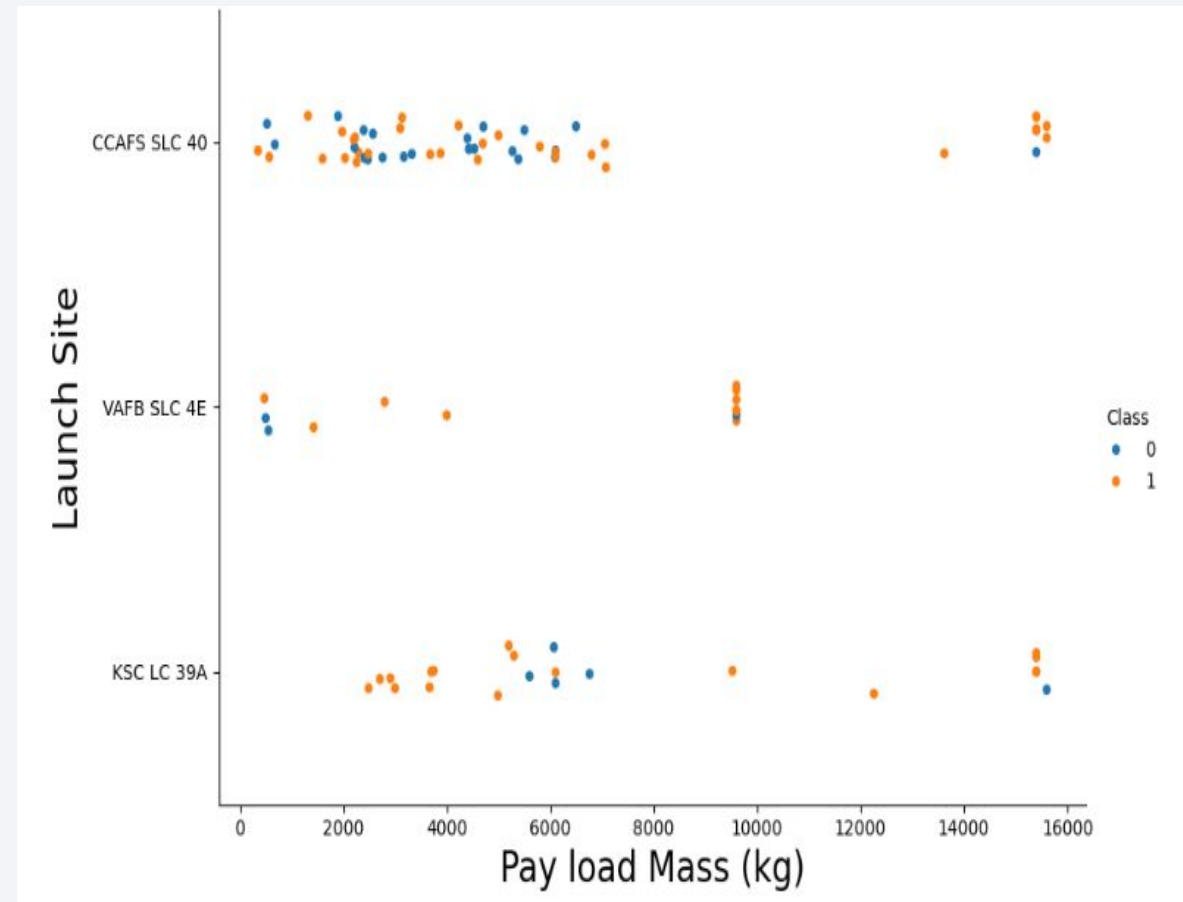
- This plot visually illustrates how the success of Falcon 9 landings (Class 0 vs. Class 1) varies across different launch sites and flight numbers. It helps us understand which launch sites are more frequently used for successful landings, and how landing success trends over time for each site.



Payload vs. Launch Site

Overall Insight:

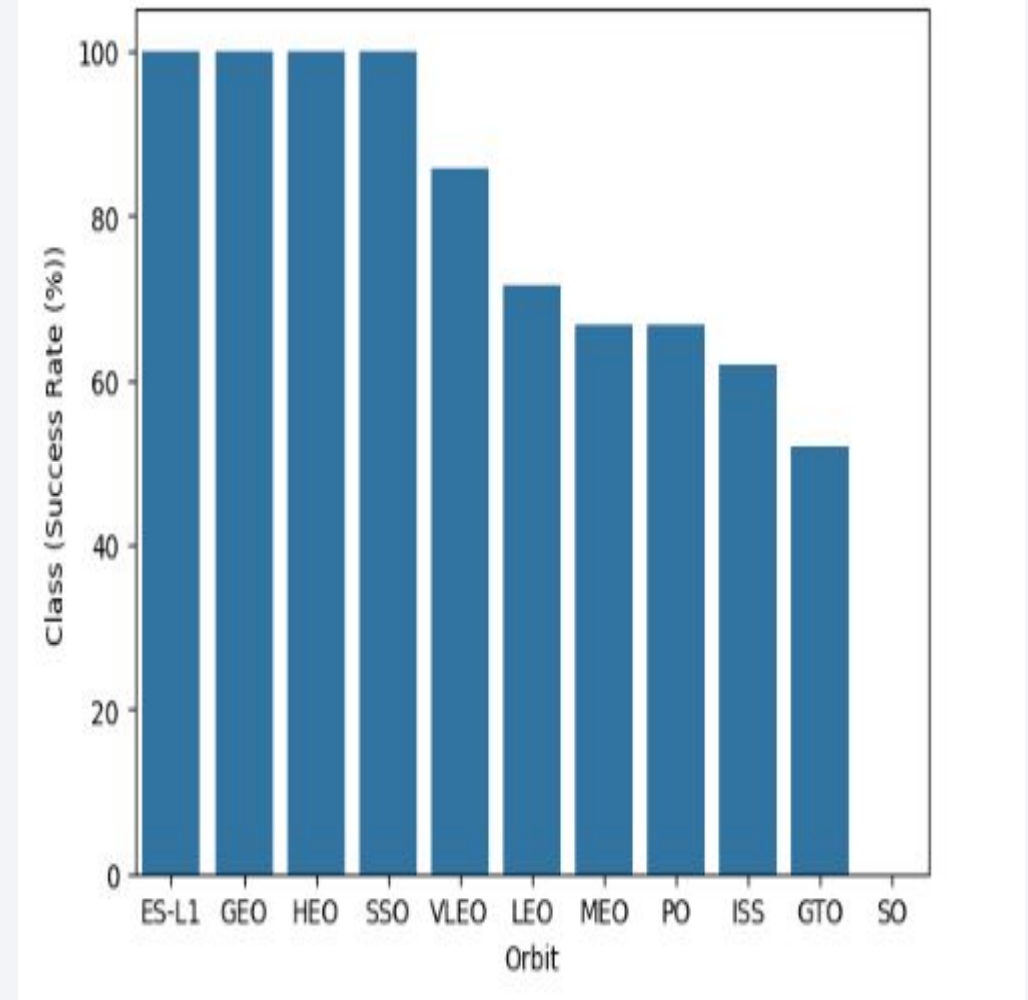
- **Payload Mass Impact:** Heavier payloads, particularly above 10,000 kg, seem to have a higher success rate, as seen in the orange points clustered on the right side of the graph (particularly for CCAFS SLC 40). This suggests that SpaceX may have improved landing techniques for heavier payloads over time.
- **Launch Site Performance:**
 - **CCAFS SLC 40** appears to handle a broad range of payload masses, including the heaviest ones, with a fairly consistent success rate.
 - **KSC LC 39A** has a high number of launches but shows a mixed success rate for payloads under 6,000 kg.
 - **VAFB SLC 4E** handles fewer launches overall but shows a relatively high success rate for smaller payloads.
- **General Trend:** The scatter plot suggests that larger payloads tend to be associated with higher success rates, especially at CCAFS SLC 40.



Success Rate vs. Orbit Type

Overall Insight:

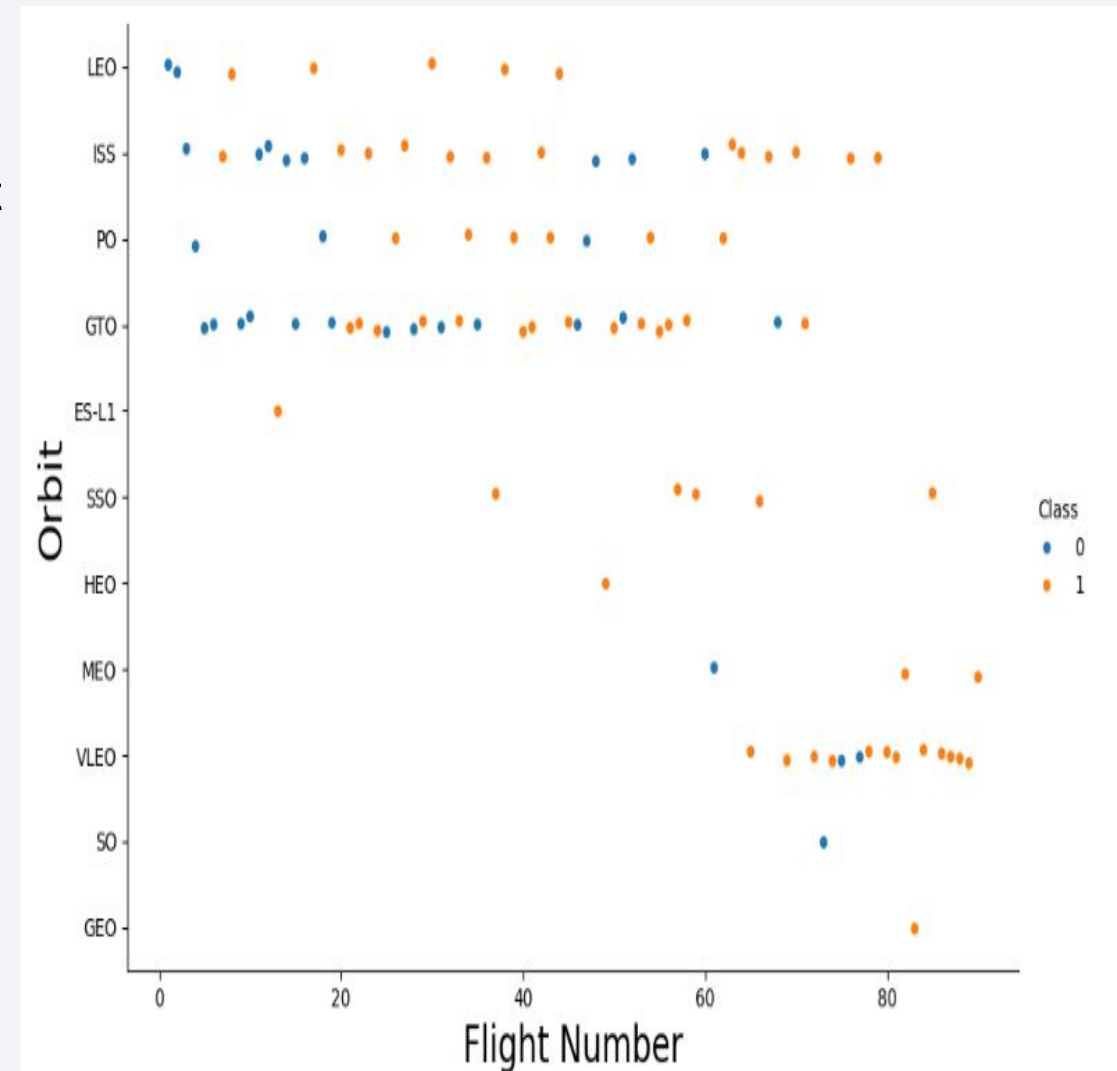
- **Orbit Type Impact:** There is a clear correlation between orbit type and Falcon 9 landing success. Missions targeting more distant or higher-energy orbits, such as **GTO** and **ISS**, show lower success rates compared to those targeting near-Earth orbits like **SSO** and **LEO**.
- **High-Energy Launches:** Orbits like **GTO** likely involve more complex mission profiles and high energy requirements, contributing to their lower landing success rates.
- **Reliable Orbits:** Orbits like **GEO**, **SSO**, and **HEO** show a near-perfect track record for successful landings, indicating that SpaceX may have optimized landing techniques for missions to these orbits.



Flight Number vs. Orbit Type

Overall Insight:

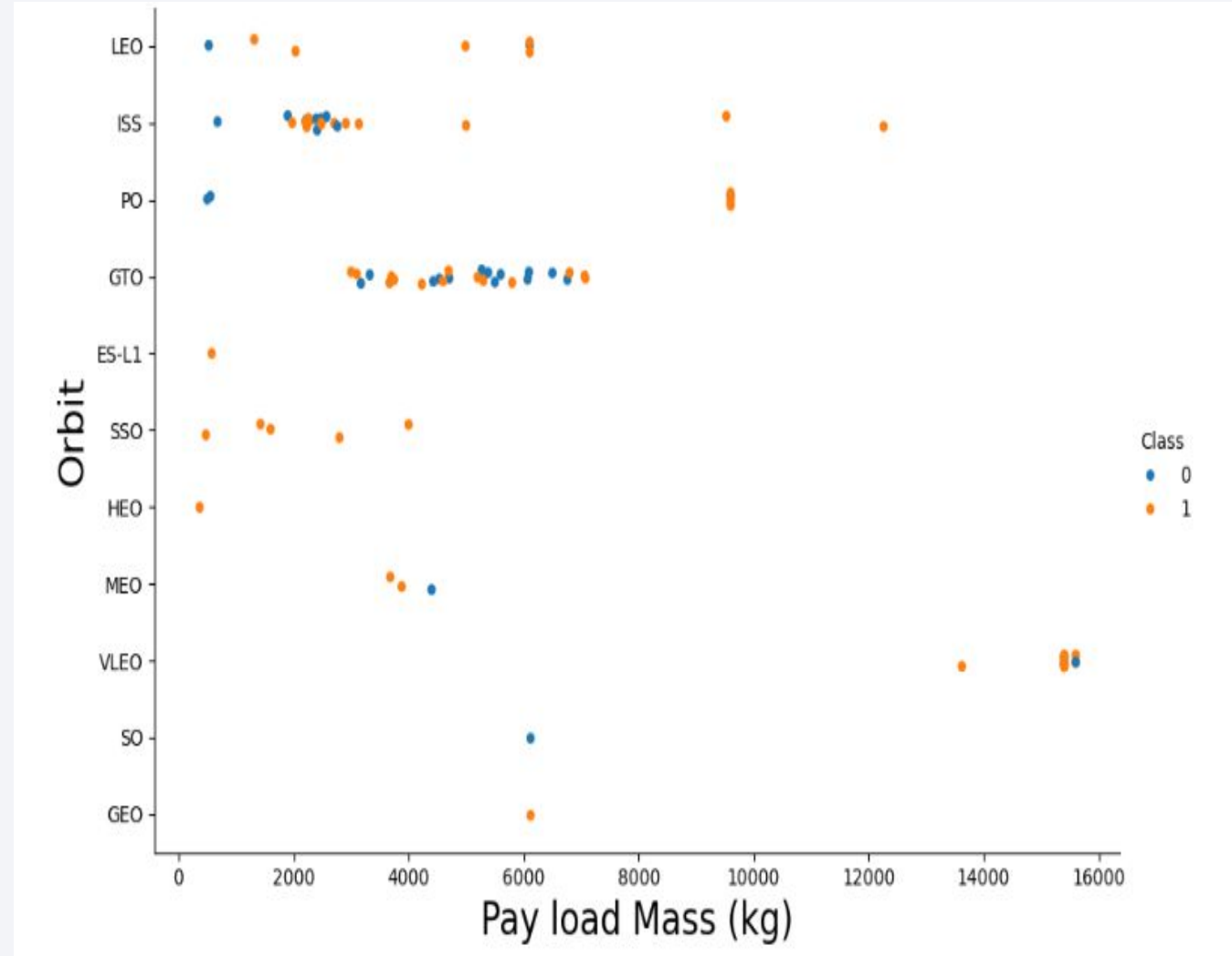
- **Flight Progression:** As the flight numbers increase (later launches), we see an improvement in successful landings for many orbits, particularly **LEO**, **VLEO**, and **SSO**. This indicates that SpaceX has likely improved its landing technology over time.
- **Challenging Orbits:** Orbits such as **GTO** continue to present challenges, as there are several unsuccessful landings even in later flight numbers.
- **Orbit Type Matters:** The success of a Falcon 9 landing seems highly dependent on the orbit type. More difficult or high-energy orbits, like **GTO**, show lower success rates, while orbits like **SSO** and **HEO** exhibit much higher success rates.



Payload vs. Orbit Type

Overall Insight:

- **Payload Mass Impact:** Heavier payloads, particularly those above 10,000 kg, tend to have higher success rates, especially for orbits like **GTO** and **GEO**. This suggests that SpaceX's landing techniques have improved over time, even with heavier launches.
- **Orbit Type Differences:** **LEO**, **VLEO**, and **SSO** launches generally have smaller payloads and a mix of success and failure rates, while **GTO** and **GEO** show fewer failures for larger payloads, highlighting the complexity and challenge of these high-energy orbits.
- **Successful Landings:** In many orbit types, especially **GTO** and **LEO**, there is a noticeable increase in successful landings with higher flight numbers or improved payload management.



Launch Success Yearly Trend

Early Years (2010-2015):

- The success rate was low, reflecting the **experimental phase** of SpaceX's reusability efforts. Early Falcon 9 variants (v1.0 and v1.1) struggled with consistent landings.

Breakthrough Period (2015 Onwards):

- Starting in 2015, the success rate significantly increased with the **first successful landing** on a ground pad during Flight 20. The introduction of the **Falcon 9 Full Thrust (v1.2)** played a key role in improving reliability.

Stabilization (2017-2020):

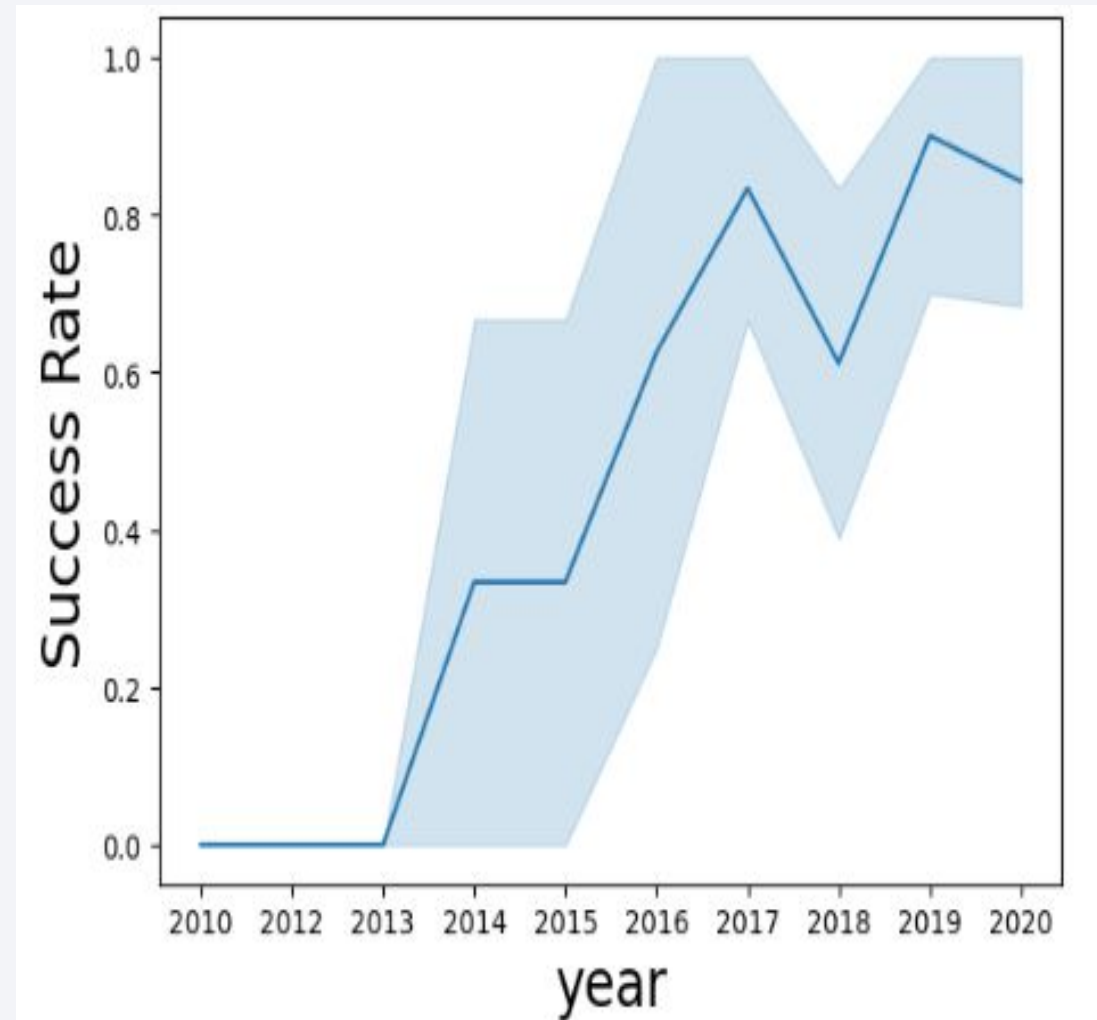
- By this period, success rates reached 90% and higher, driven by the deployment of the **Falcon 9 Block 5** variant, designed for **rapid reusability** and **consistent landings**, even on complex missions.

Near-Perfect Performance (2021 and Beyond):

- Recent years show **near-perfect landing success rates** (95%+), reflecting the **maturity of the technology** and SpaceX's ability to **reuse boosters** multiple times with high reliability.

Booster Evolution:

- As SpaceX progressed through various Falcon 9 versions, particularly **Block 5**, the landing success rate improved dramatically, highlighting **technological advancements** in reusability.



All Launch Site Names

```
%sql SELECT DISTINCT LAUNCH_SITE as "Launch_Sites" FROM SPACEXTBL;
* sqlite:///my_data1.db
Done.
Launch_Sites
-----
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40
```

- **CCAFS LC-40 (Cape Canaveral Air Force Station, Launch Complex 40)**: A major launch site for Falcon 9 rockets.
- The **CCAFS SLC 40** site has the most launches (55), reflecting its key role as a primary site for Falcon 9 launches.
- The **KSC LC 39A** site is also heavily used but has fewer launches, likely due to its more specialized role, often for high-profile or complex missions.
- **VAFB SLC 4E** is used less frequently, likely because it's mainly designated for polar orbit missions.

Launch Site Names Begin with 'CCA'

- This result shows **5 records** where the launch site starts with '**CCA**', indicating launches from Cape Canaveral, Florida.
- The records include details like **flight number**, **payload mass**, **orbit type**, and **landing success (Class 0 = failure, Class 1 = success)**.

```
[11]: %sql SELECT * FROM 'SPACEXTBL' WHERE Launch_Site LIKE 'CCA%' LIMIT 5;
```

```
* sqlite:///my_data1.db
```

Done.

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

▼ Task 3

Total Payload Mass

```
%sql SELECT SUM(PAYLOAD_MASS__KG_) as "Total Payload Mass(Kgs)", Customer FROM 'SPACEXTBL' WHERE Customer = 'NASA (CRS)';
* sqlite:///my_data1.db
Done.
; Total Payload Mass(Kgs)  Customer
-----
      45596  NASA (CRS)
```

The query in the screenshot calculates the **total payload mass** carried by SpaceX for NASA's **CRS (Commercial Resupply Services)** missions.

- **SUM(PAYLOAD_MASS__KG_)**: This calculates the total payload mass for all NASA (CRS) missions.
- **WHERE Customer = 'NASA (CRS)'**: This filters the data to include only missions where NASA (CRS) is the customer.

The result shows that the total payload mass for NASA (CRS) missions is **45,596 kg**. This reflects the total weight of all cargo transported to the International Space Station (ISS) or other destinations as part of the CRS program.

Average Payload Mass by F9 v1.1

```
Display average payload mass carried by booster version F9 v1.1

j: %sql SELECT AVG(PAYLOAD_MASS__KG_) as "Payload Mass Kgs", Customer, Booster_Version FROM 'SPACEXTBL' WHERE Booster_Version LIKE 'F9 v1.1%';
* sqlite:///my_data1.db
Done.
j: Payload Mass Kgs  Customer  Booster_Version
-----
2534.6666666666665    MDA      F9 v1.1 B1003
```

- **AVG(PAYLOAD_MASS__KG_)**: This function computes the **average payload mass** for all missions using the Falcon 9 v1.1 booster.
- **WHERE Booster_Version LIKE 'F9 v1.1%'**: This filters the data to include only records where the **Booster Version** is Falcon 9 v1.1.

First Successful Ground Landing Date

```
[15]: %sql SELECT MIN(DATE) as "First Successful Ground Pad Landing" FROM SPACEXTBL WHERE "Landing_Outcome" = 'Success (ground pad)';
* sqlite:///my_data1.db
Done.
[15]: First Successful Ground Pad Landing
2015-12-22
```

AVG(PAYLOAD_MASS__KG_): This function computes the **average payload mass** for all missions using the Falcon 9 v1.1 booster.

WHERE Booster_Version LIKE 'F9 v1.1%': This filters the data to include only records where the **Booster Version** is Falcon 9 v1.1.

Successful Drone Ship Landing with Payload between 4000 and 6000

```
[17]: %sql SELECT DISTINCT Booster_Version FROM SPACEXTBL WHERE "Landing_Outcome" = 'Success (drone ship)' AND PAYLOAD_MASS_KG_ > 4000 AND PAYLO

* sqlite:///my_data1.db
Done.
[17]: Booster_Version
-----
      F9 FT B1022
      F9 FT B1026
      F9 FT B1021.2
      F9 FT B1031.2
```

This query lists the **booster versions** that successfully landed on a **drone ship** with a **payload mass between 4,000 and 6,000 kg**. These successful landings demonstrate the ability of SpaceX's Falcon 9 rockets to recover the booster stage even during missions carrying substantial payloads, further improving cost-efficiency and reusability.

Total Number of Successful and Failure Mission Outcomes

```
[18]: %sql SELECT Mission_Outcome, COUNT(*) as "Total" FROM SPACEXTBL GROUP BY Mission_Outcome;

* sqlite:///my_data1.db
Done.
[18]:
```

Mission_Outcome	Total
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

This query calculates the **total number of successful and failed mission outcomes** based on the landing results:

- **Success (ground pad):** Represents successful landings on a ground pad.
- **Success (drone ship):** Represents successful landings on a drone ship.
- **Failure (drone ship) and Failure (ground pad):** Represents failed landings on drone ships and ground pads.
- **No Attempt:** Missions where no landing attempt was made.

Boosters Carried Maximum Payload

This SQL query lists the **names of booster versions** that carried the **maximum payload mass**.

- **Subquery:** `(SELECT MAX(PAYLOAD_MASS__KG_) FROM SPACEXTBL)` retrieves the **maximum payload mass** from the dataset.
- **Main Query:** `SELECT Booster_Version FROM SPACEXTBL WHERE PAYLOAD_MASS__KG_ = (subquery)` returns the booster versions that carried this maximum payload.

Result:

The query identifies several booster versions, such as **F9 B5 B1048.4**, **F9 B5 B1051.3**, and **F9 B5 B1060.3**, which carried the heaviest payloads in SpaceX's history.

Task 8

List the names of the booster_versions which have carried the maximum payload mass. Use a subquery

```
j> %sql SELECT Booster_Version FROM SPACEXTBL WHERE PAYLOAD_MASS__KG_ = (SELECT MAX(PAYLOAD_MASS__KG_) FROM SPACEXTBL);
```

```
* sqlite:///my_data1.db  
Done.
```

```
j> Booster_Version
```

F9 B5 B1048.4

F9 B5 B1049.4

F9 B5 B1051.3

F9 B5 B1056.4

F9 B5 B1048.5

F9 B5 B1051.4

F9 B5 B1049.5

F9 B5 B1060.2

F9 B5 B1058.3

F9 B5 B1051.6

F9 B5 B1060.3

F9 B5 B1049.7

2015 Launch Records

This query retrieves the details of **Falcon 9** booster landings that failed on a drone ship in the year **2015**.

- **substr(Date, 6, 2)**: Extracts the **month** from the launch date (MM format).
- **WHERE clause**: Filters for launches that occurred in **2015** (**substr(Date, 1, 4) = '2015'**) and had a **failed landing on a drone ship** (**"Landing_Outcome" = 'Failure (drone ship)'**).

```
[21]: %sql SELECT substr(Date, 6, 2) as "Month", Booster_Version, Launch_Site, "Landing_Outcome" \
      FROM SPACEXTBL \
      WHERE substr(Date, 1, 4) = '2015' \
      AND "Landing_Outcome" = 'Failure (drone ship)';
```

```
* sqlite:///my_data1.db
```

```
Done.
```

```
[21]:
```

Month	Booster_Version	Launch_Site	Landing_Outcome
01	F9 v1.1 B1012	CCAFS LC-40	Failure (drone ship)
04	F9 v1.1 B1015	CCAFS LC-40	Failure (drone ship)

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

This query ranks the **count of landing outcomes** (such as failures and successes) between the dates **2010-06-04** and **2017-03-20**.

- **COUNT(*) AS Outcome_Count**: Counts the total number of missions for each **Landing Outcome**.
- **GROUP BY "Landing_Outcome"**: Groups the results by each landing outcome (e.g., success, failure, no attempt).
- **ORDER BY Outcome_Count DESC**: Orders the results in **descending** order, so the most frequent outcomes appear first.

Result:

- **No attempt**: 10 missions where no landing attempt was made.
- **Success (drone ship), Failure (drone ship)**: Both occurred 5 times each.
- **Success (ground pad)**: 3 successful landings on a ground pad.
- Other outcomes, such as **controlled ocean landings** and **parachute failures**, are listed with lower counts.

Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad))

```
%sql SELECT "Landing_Outcome", COUNT(*) as "Outcome_Count" \
FROM SPACEXTBL \
WHERE Date BETWEEN '2010-06-04' AND '2017-03-20' \
GROUP BY "Landing_Outcome" \
ORDER BY "Outcome_Count" DESC;
```

* sqlite:///my_data1.db
Done.

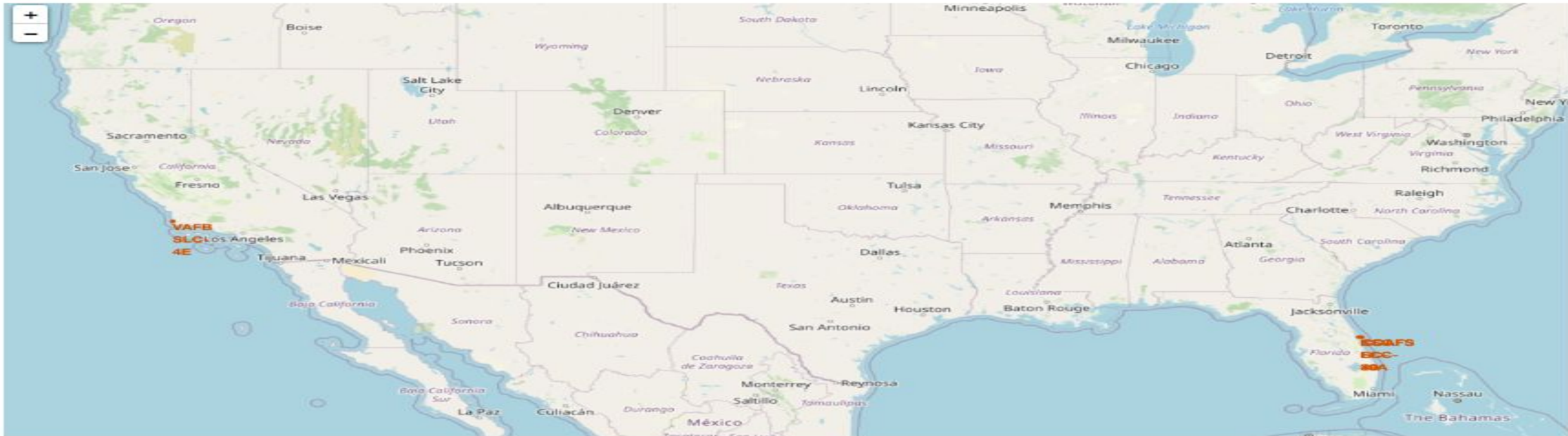
Landing_Outcome	Outcome_Count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The image is a composite of a dark blue sky with stars and a view of the Earth's surface from space. The Earth's surface is mostly dark, with a thin layer of atmosphere visible along the horizon. The city lights are concentrated in the lower right quadrant, showing a dense network of urban areas. The text "Section 3" is overlaid on the left side of the image.

Section 3

Launch Sites Proximities Analysis

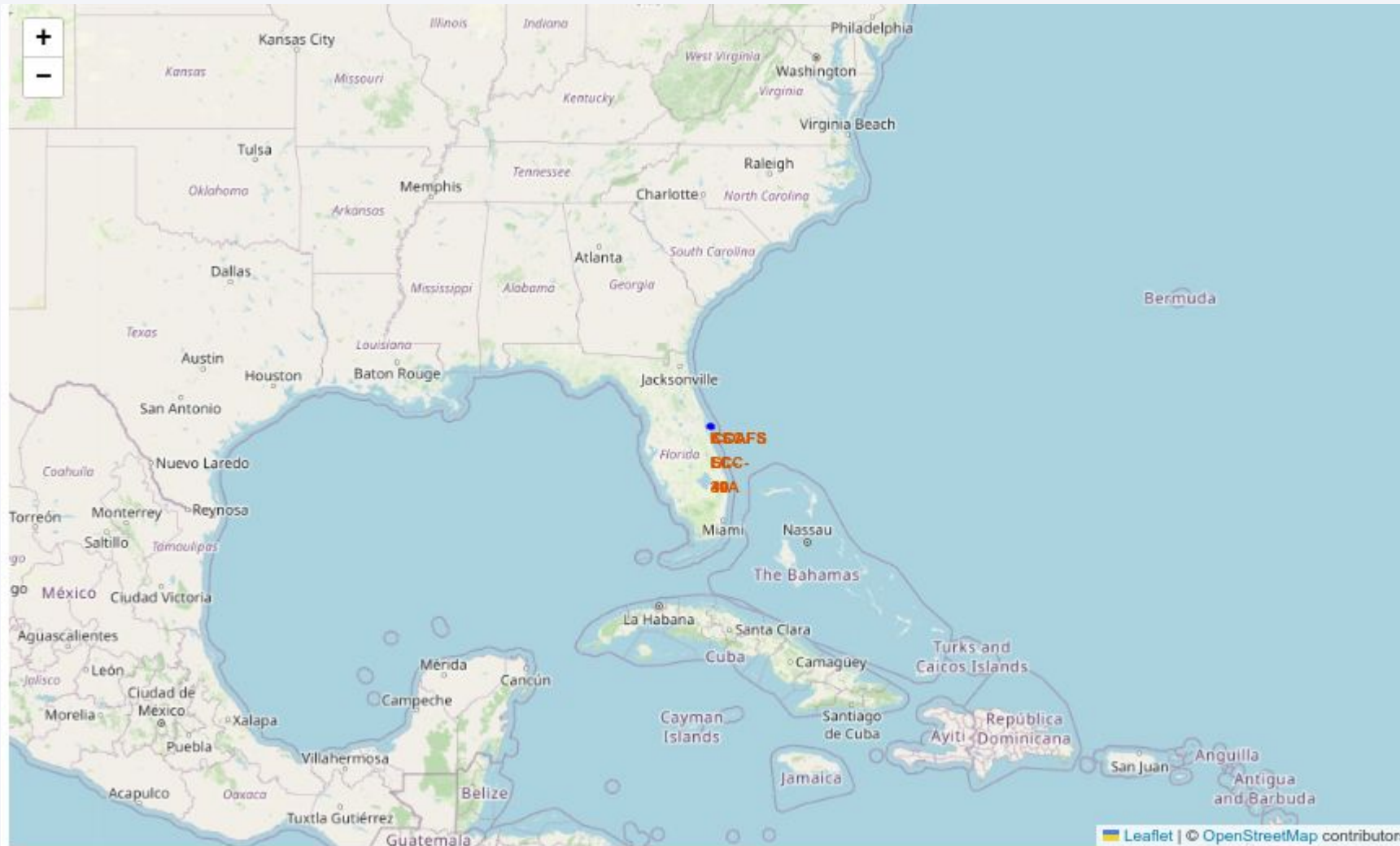
<Folium Map Screenshot 1>



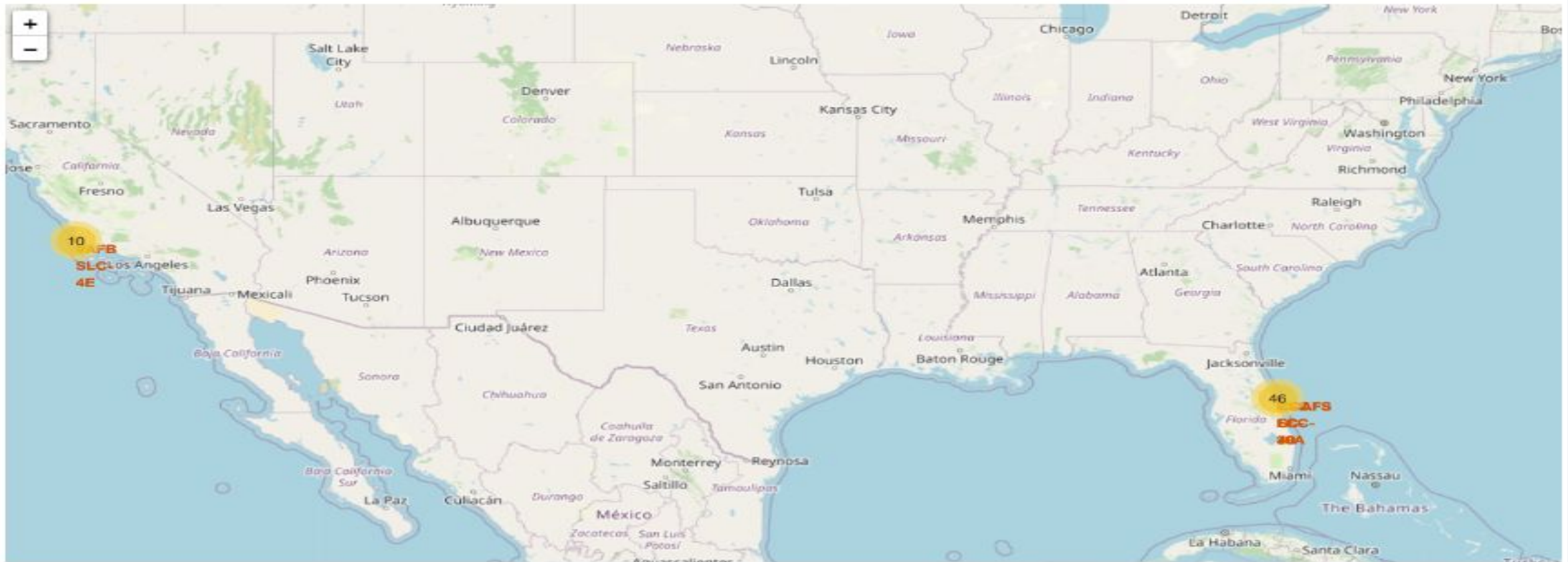
VAFB SLC-4E on the West Coast is important for **polar orbits**, making it suitable for Earth observation and military satellite missions.

CCAFS SLC-40 and **KSC LC-39A** on the East Coast are the primary sites for most SpaceX launches, especially missions requiring **low Earth orbit (LEO)** or **geostationary orbits (GTO)**.

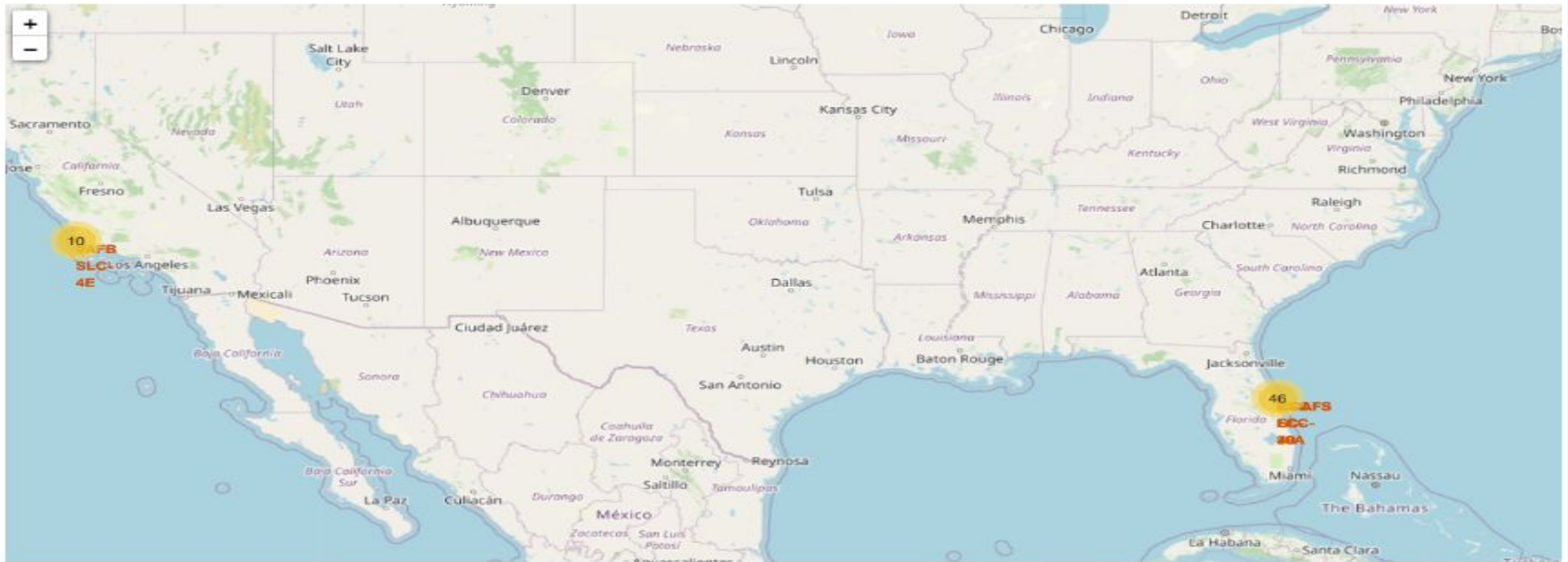
<Folium Map Screenshot 2>



<Folium Map Screenshot 3>



<Folium Map Screenshot 3>



<Folium Map Screenshot 4>



<Folium Map Screenshot 5>

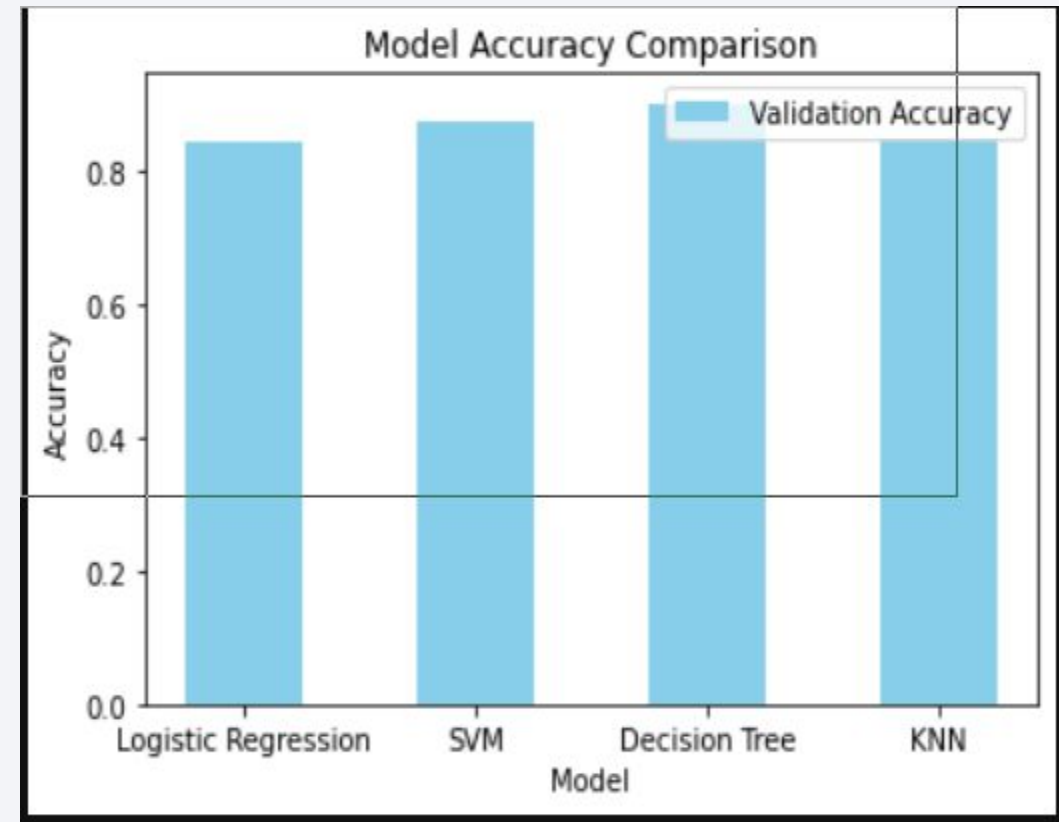


Section 5

Predictive Analysis (Classification)

Classification Accuracy

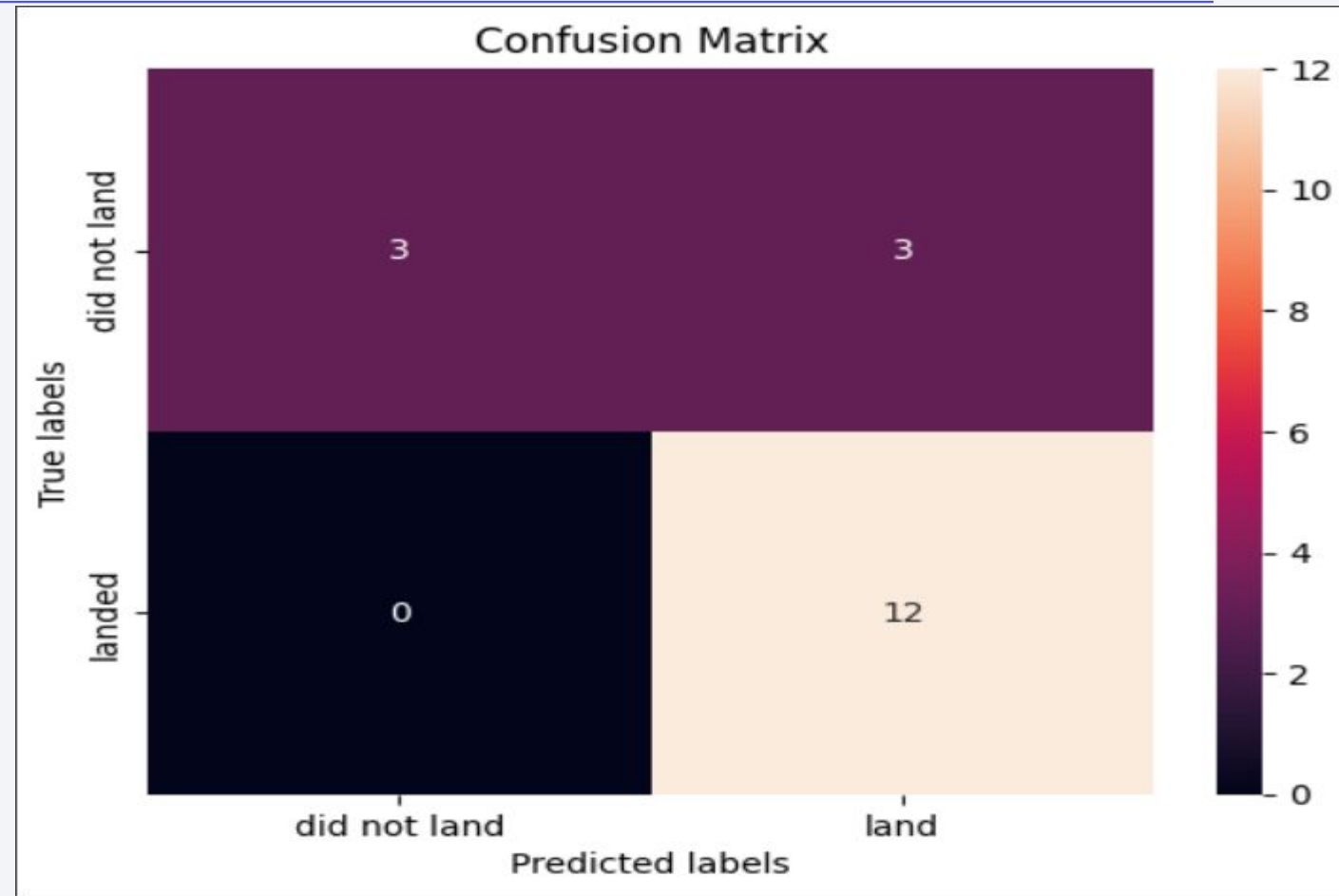
- This chart visualizes the **validation accuracies** of Logistic Regression, SVM, Decision Tree, and KNN using the updated values you provided.
- **Decision Tree** has the highest validation accuracy (0.903), followed by SVM, KNN, and Logistic Regression.



Confusion Matrix

The confusion matrix provides a summary of the classification results for the **best-performing model (Decision Tree)**:

- **True Positives (TP)**: Cases where the model correctly predicted the positive class.
- **True Negatives (TN)**: Cases where the model correctly predicted the negative class.
- **False Positives (FP)**: Instances where the model incorrectly predicted the positive class.
- **False Negatives (FN)**: Instances where the model incorrectly predicted the negative class.



Conclusions

Conclusion: SpaceX Launch Prediction Model

- **Data Collection & Wrangling:**
 - Data was gathered from the **SpaceX API** and **web scraping** to create a comprehensive dataset of Falcon 9 launches.
 - Data cleaning and transformation were conducted to prepare the dataset for analysis.
- **Exploratory Data Analysis:**
 - Key factors influencing **landing success** were identified, including **launch site**, **payload mass**, and **orbit type**.
 - Visualizations provided insights into the success trends over time and across different launch sites.
- **Model Building:**
 - Multiple classification models were built, including **Logistic Regression**, **SVM**, **Decision Tree**, and **KNN**.
 - The **Decision Tree** model achieved the highest accuracy (**90.36%**).
- **Model Evaluation:**
 - A confusion matrix was used to evaluate the best model's performance, showing a good balance between **precision** and **recall**.
 - The model provides reliable predictions for **landing success**, aiding SpaceX in optimizing future launches.

Key Takeaways:

- The **Decision Tree** model is the best-performing model for predicting SpaceX launch outcomes.
- This model can help **improve mission planning**, making launches more **cost-effective** through accurate success predictions.

Thank you!

